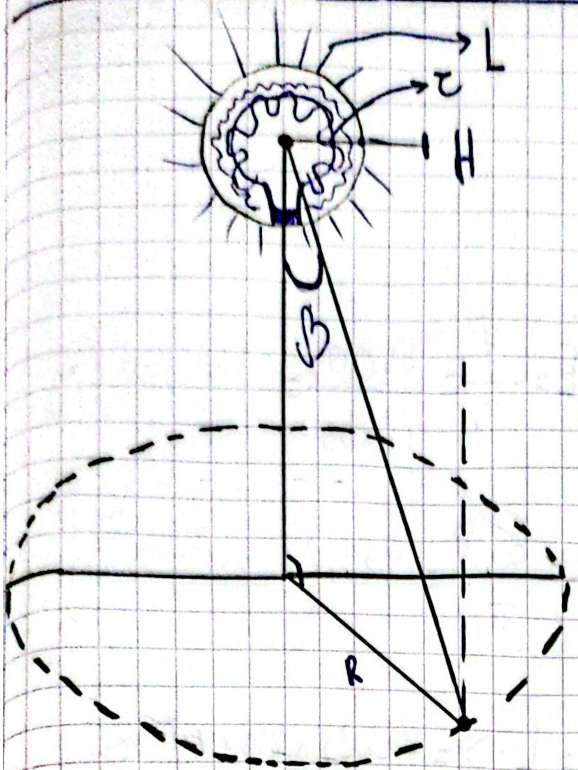
Homework 4 | Mirjabbar Badalov | MPA-SVT



$D = 0{,}65m \Rightarrow r = 0{,}325\,m$

$L = 3100\,cd \cdot m^{-2}$

$H = 3{,}2\,m$

$R = 2{,}2\,m$

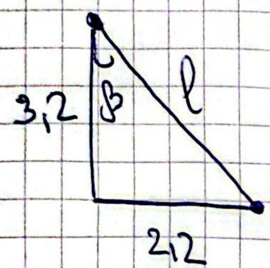
Problem 7

① Luminous intensity:

$$L = \frac{I_0}{S} \quad I_0 = L \cdot S = 3100 \cdot \pi \cdot 0{,}325^2 \Rightarrow$$

$$\left[I_0 = 1028{,}675\ cd \right]$$

② For normal illuminance on the point we have to define angle and distance from source to the point:



$$l = \sqrt{3{,}2^2 + 2{,}2^2} = 3{,}88\,m$$

$$\cos\beta = \frac{3{,}2}{3{,}88} = 0{,}825$$

Then normal illuminance will be:

$$E_n = \frac{I_0}{l^2} \cdot \cos\beta = \frac{1028{,}675}{3{,}88^2} \cdot 0{,}825 \Rightarrow$$

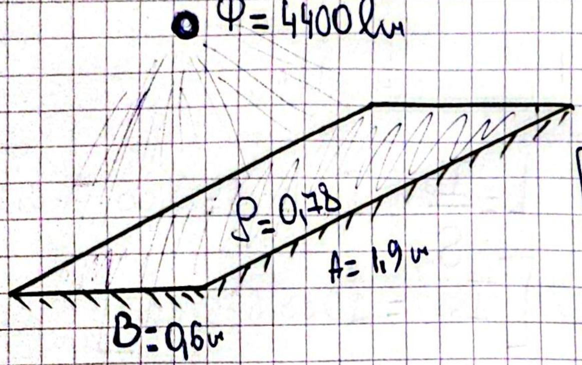
$$\left[E_n = 56{,}37\ lx \right]$$

Problem 8

A lambert surface size [A x B], Reflectance : [R] has illuminated by a luminous flux [Flm]

Illumination E = ?

Luminance = ? Luminous intensity = ? I_0 = ?



$$\left[E = \frac{\Phi}{S} = \frac{4400}{1,9 \times 0,6} = 3859,65\ lx\right]$$

$$\left[L = \frac{E \cdot \rho}{\pi} = 958,28\ cd \cdot m^{-2}\right]$$

$$I_0 = L \cdot S = 958,28 \times 0,6 \times 1,9 \Rightarrow$$

$$\left[I_0 = 1092,44\ cd\right]$$